Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 12: Mesh Current Analysis

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Colour code used in these notes and in the slides

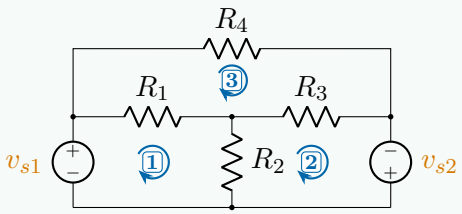
■ current i, mesh current i_{Mk} around mesh k ■ voltage v, node voltage e ■ supermesh Ⓢ

Motivation. Node voltage analysis takes the $n-1$ node voltages as the unknowns. Then KVL holds automatically, and we write KCL. **Mesh current analysis** does the opposite. Its unknowns are the $\ell = b-n+1$ mesh currents. Every branch current is $\mathbf{i} = M^T \mathbf{i}_M$, so KCL holds automatically (previously), and we write KVL around each inner mesh. Meshes need a planar circuit: one drawn with no crossing branches.

1. The procedure, on an example

For now, the circuit has only resistors and independent voltage sources; current sources and dependent sources come later. The circuit of Example 12.1 has $b = 6$ branches, so 12 branch voltages and currents. Mesh current analysis needs only 3 unknowns. Four steps find them.

Example 12.1 — mesh current analysis step by step



1. **Inner meshes.** The circuit is planar; its windows are the inner meshes, $\ell = 6 - 4 + 1 = 3$.
2. **Mesh currents.** Give each a clockwise mesh current i_{M1}, i_{M2}, i_{M3}. Every branch current is a combination of them ($i_{R_2} = i_{M1} - i_{M2}$ downwards), so KCL holds automatically.
3. **Voltages.** Along mesh k, a resistor R shared with mesh j has the voltage $R(i_{Mk} - i_{Mj})$; on mesh k only, $R\, i_{Mk}$.
4. **KVL** around every inner mesh:

$$\begin{aligned}
\text{mesh 1:} \quad & -v_{s1} + R_1(i_{M1} - i_{M3}) + R_2(i_{M1} - i_{M2}) = 0 \\
\text{mesh 2:} \quad & R_2(i_{M2} - i_{M1}) + R_3(i_{M2} - i_{M3}) - v_{s2} = 0 \\
\text{mesh 3:} \quad & R_4\, i_{M3} + R_3(i_{M3} - i_{M2}) + R_1(i_{M3} - i_{M1}) = 0
\end{aligned}$$

Mesh 1 meets v_{s1} from $-$ to $+$, so it enters with a minus sign. Three equations, three unknowns; then Steps 2 and 3 give every branch current and voltage.

In general, $\mathbf{i} = M^T \mathbf{i}_M$ (KCL holds automatically), the element laws give $\mathbf{v}$, and KVL $M\mathbf{v} = \mathbf{0}$ gives ℓ equations in the ℓ mesh currents.

2. Matrix form

With all mesh currents clockwise, the three equations of Example 12.1 become

$$\underbrace{\begin{bmatrix} R_1+R_2 & -R_2 & -R_1 \\ -R_2 & R_2+R_3 & -R_3 \\ -R_1 & -R_3 & R_1+R_3+R_4 \end{bmatrix}}_{Z_M} \underbrace{\begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix}}_{\mathbf{i}_M} = \underbrace{\begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix}}_{\mathbf{v}_M}.$$

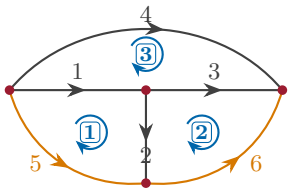
Z_M is the **mesh impedance matrix**, and $Z_M \mathbf{i}_M = \mathbf{v}_M$ are the **mesh equations**; $\mathbf{v}_M$ holds the source voltages rising along each mesh. For a circuit of resistors and voltage sources, they can be written down directly.

Writing Z_M directly from the circuit: three rules

- **Diagonal** entry (k,k): the sum of the resistances around mesh k.
- **Off-diagonal** entry (k,j): minus the sum of the resistances shared by meshes k and j.
- **Right side**, row k: the sum of the source voltages rising (met from $-$ to $+$) along mesh k.

So Z_M is **symmetric**: the entry (k,j) equals the entry (j,k).

Z_M from the graph. In the graph of Example 12.1, branches 1–4 are the resistors $R_1,\dots,R_4$, and branches 5 and 6 are the sources v_{s1}, v_{s2}; a source branch points from $+$ to $-$. The mesh matrix M has one row for each inner mesh; an entry is $+1$ if the branch goes along the mesh and -1 if against it.



$$M = \begin{array}{c} \\ \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \left[\begin{array}{cccc|cc} 1 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 1 & 0 & 0 & -1 \\ -1 & 0 & -1 & 1 & 0 & 0 \end{array}\right] \end{array}$$

M_R: resistors M_S: sources

KVL $M\mathbf{v} = \mathbf{0}$ reads $M_R\mathbf{v}_R + M_S\mathbf{v}_S = \mathbf{0}$. The resistors obey $\mathbf{v}_R = R\,\mathbf{i}_R$ with $R = \mathrm{diag}(R_1,\dots,R_4)$. KCL, $\mathbf{i} = M^T\mathbf{i}_M$, splits into rows: $\begin{bmatrix} \mathbf{i}_R \\ \mathbf{i}_S \end{bmatrix} = \begin{bmatrix} M_R^T \\ M_S^T \end{bmatrix} \mathbf{i}_M$, so $\mathbf{i}_R = M_R^T\mathbf{i}_M$ (resistor rows) and $\mathbf{i}_S = M_S^T\mathbf{i}_M$ (source rows); only the resistor rows are needed. In Example 12.1, M_R^T is the resistor columns 1–4 of M written as rows, and

$$\mathbf{i}_R = M_R^T\mathbf{i}_M = \begin{bmatrix} 1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} i_{M1} - i_{M3} & i_{M1} - i_{M2} & i_{M2} - i_{M3} & i_{M3} \end{bmatrix}^T :$$

each resistor current is the mesh currents along its arrow minus those against it. Together,

$$M_R\,R\,M_R^T\,\mathbf{i}_M = -M_S\mathbf{v}_S \quad \Rightarrow \quad Z_M = M_R\,R\,M_R^T, \qquad \mathbf{v}_M = -M_S\mathbf{v}_S.$$

For Example 12.1,

$$M_R\,R = \begin{bmatrix} R_1 & R_2 & 0 & 0 \\ 0 & -R_2 & R_3 & 0 \\ -R_1 & 0 & -R_3 & R_4 \end{bmatrix}, \qquad -M_S\mathbf{v}_S = -\begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_{s1} \\ v_{s2} \end{bmatrix} = \begin{bmatrix} v_{s1} \\ v_{s2} \\ 0 \end{bmatrix},$$

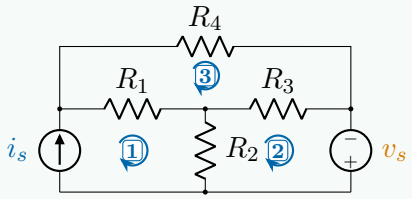
and $M_R\,R\,M_R^T$ is exactly the Z_M above: the three rules read this product off entry by entry. Finally, $Z_M^T = M_R R^T M_R^T = Z_M$, since R is diagonal: with resistors and voltage sources, Z_M is always symmetric.

3. A current source on one mesh only

A current source on one mesh only

If an independent current source i_s is on mesh k only (on the outer boundary), then $i_{Mk} = \pm i_s$ is known. So i_{Mk} is not an unknown, and KVL is not written around mesh k: the voltage of the current source is not known.

Example 12.2 — a current source on mesh 1



The source i_s points along mesh 1, so $i_{M1} = i_s$; the unknowns are only i_{M2}, i_{M3}.

mesh 2: $R_2(i_{M2} - i_s) + R_3(i_{M2} - i_{M3}) - v_s = 0$

mesh 3: $R_1(i_{M3} - i_s) + R_3(i_{M3} - i_{M2}) + R_4\, i_{M3} = 0$

In matrix form, Example 12.2 reads

$$\begin{bmatrix} R_2 + R_3 & -R_3 \\ -R_3 & R_1 + R_3 + R_4 \end{bmatrix} \begin{bmatrix} i_{M2} \\ i_{M3} \end{bmatrix} = \begin{bmatrix} v_s + R_2 i_s \\ R_1 i_s \end{bmatrix}.$$

The known i_s moves to the right side: $R_2 i_s$ acts like a source voltage on mesh 2.

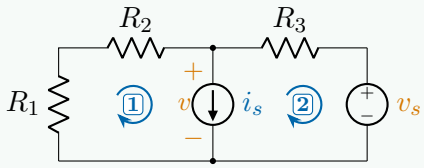
4. A current source between two meshes: modified mesh analysis

If a current source (independent or dependent) is shared by two meshes, KVL around either mesh needs its voltage, but its voltage cannot be written with mesh currents. Its law fixes its current, not its voltage. **Modified mesh current analysis** offers two ways out.

Method I

Let the unknown voltage v of the current source be one more unknown. Write KVL around every inner mesh, and add the law of the source as a **constraint equation**.

Example 12.3 — Method I



mesh 1: $(R_1 + R_2)\, i_{M1} + v = 0$

mesh 2: $R_3\, i_{M2} + v_s - v = 0$

constraint: $i_{M1} - i_{M2} = i_s$

$$\begin{bmatrix} R_1 + R_2 & 0 & 1 \\ 0 & R_3 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ -v_s \\ i_s \end{bmatrix}$$

Three unknowns and three equations; the last row is the constraint.

Method II: the supermesh. Add the KVL equations of meshes 1 and 2: v enters mesh 1 with $+$ and mesh 2 with $-$, so it cancels. The sum is KVL around the outer loop of the two meshes, which avoids the current source.

Method II: the supermesh

Write KVL around two meshes as if they were a single mesh, skipping the current source between them. Such a loop is a **supermesh**. Add the law of the source as a constraint equation.

Example 12.3, revisited with a supermesh (circuit above, the loop Ⓢ around meshes 1 and 2): the unknowns are i_{M1}, i_{M2}, and

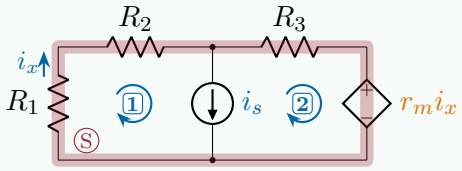
$$\left.\begin{array}{ll} \text{supermesh:} & (R_1 + R_2)\, i_{M1} + R_3\, i_{M2} + v_s = 0 \\ \text{constraint:} & i_{M1} - i_{M2} = i_s \end{array}\right\} \quad \Rightarrow \quad \begin{bmatrix} R_1 + R_2 & R_3 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix} = \begin{bmatrix} -v_s \\ i_s \end{bmatrix}.$$

Method II has one unknown fewer than Method I and gives the same mesh currents; in both, the matrix is no longer symmetric.

5. Dependent sources

A dependent source is handled like an independent one of the same kind. Its control variable is then written with mesh currents; this gives one more constraint equation.

Example 12.4 — a dependent voltage source



Question. Write the mesh equations.
Answer. Treat the dependent source as a voltage source $r_m i_x$. The current source is between meshes 1 and 2, so use the supermesh Ⓢ:

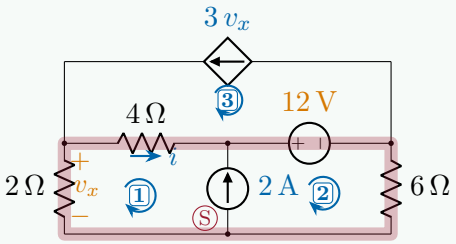
$$\text{supermesh:}\quad (R_1+R_2)\,i_{M1}+R_3\,i_{M2}+r_m i_x=0$$
$$\text{constraint:}\quad i_{M1}-i_{M2}=i_s$$

The control is $i_x = i_{M1}$.

$$\begin{bmatrix} R_1+R_2+r_m & R_3 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} i_{M1} \\ i_{M2} \end{bmatrix}=\begin{bmatrix} 0 \\ i_s \end{bmatrix} \quad\Rightarrow\quad i_{M1}=\frac{R_3\,i_s}{R_1+R_2+R_3+r_m},\quad i_{M2}=i_{M1}-i_s.$$

With $i_x = i_{M1}$, the term $r_m i_{M1}$ moves to the left: the dependent source changes one entry of the matrix.

Example 12.5 — a dependent current source



Question. Find i, the current of the 4 Ω.
Answer. The 2 A source is between meshes 1 and 2: use the supermesh Ⓢ. The dependent source is on mesh 3 only: it fixes i_{M3}. So KVL is written around Ⓢ only:

$$2\,i_{M1}+4(i_{M1}-i_{M3})+12+6\,i_{M2}=0 \qquad (1)$$

The 2 A source gives $i_{M2}-i_{M1}=2$ (2). The dependent source gives $i_{M3}=-3\,v_x$; i_{M1} flows up through the 2 Ω, from − to +, so $v_x=-2\,i_{M1}$ and $i_{M3}=6\,i_{M1}$ (3). In matrix form,

$$\begin{bmatrix} 6 & 6 & -4 \\ -1 & 1 & 0 \\ -6 & 0 & 1 \end{bmatrix}\begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix}=\begin{bmatrix} -12 \\ 2 \\ 0 \end{bmatrix} \quad\Rightarrow\quad i_{M1}=2\,\text{A},\quad i_{M2}=4\,\text{A},\quad i_{M3}=12\,\text{A},$$

so $i=i_{M1}-i_{M3}=2-12=-10\,\text{A}$.

Remark: a dependent current source on one mesh only

If a dependent current source is on mesh k only, do not write KVL around mesh k: its voltage is not known. Write its constraint instead.

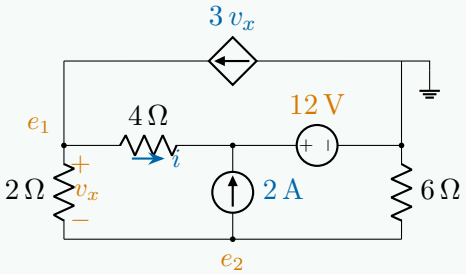
6. Node or mesh?

Which method needs fewer equations for Example 12.5? Count before solving. With mesh currents there are $\ell = 3$ unknowns, and each current source fixes one of them: $3-2=1$ KVL equation is left. With node voltages there are $n-1=3$ unknowns, and each voltage source fixes one of them: $3-1=2$ KCL equations are left.

Tip. Count before you start: for mesh currents, ℓ minus the current sources; for node voltages, $n-1$ minus the voltage sources. Mesh current analysis needs a planar circuit; node voltage analysis always works.

To check, solve Example 12.5 with node voltages. Take the right node as the ground. The 12 V source goes from the middle node to the ground, so that node is at 12 V; the unknowns are e_1 and e_2.

Example 12.5 with node voltages



$$\text{node 1:}\quad \frac{e_1-e_2}{2}+\frac{e_1-12}{4}-3\,v_x=0$$

$$\text{node 2:}\quad 2+\frac{e_2-e_1}{2}+\frac{e_2}{6}=0$$

$$\text{constraint:}\quad v_x=e_1-e_2$$

$$\begin{bmatrix}-\frac{9}{4} & \frac{5}{2}\\ -\frac{1}{2} & \frac{2}{3}\end{bmatrix}\begin{bmatrix}e_1\\ e_2\end{bmatrix}=\begin{bmatrix}3\\ -2\end{bmatrix}\ \Rightarrow\ e_1=-28\,\text{V},\ e_2=-24\,\text{V},\quad i=\frac{e_1-12}{4}=-10\,\text{A},$$

the same answer.

Summary

In the circuit	What to do
resistors and voltage sources	KVL around every inner mesh: $Z_M\mathbf{i}_M=\mathbf{v}_M$, from three simple rules; $Z_M=M_RRM_R^T$ is symmetric
a current source on one mesh only	i_{Mk} is known; no KVL around mesh k
a current source between meshes	Method I: its voltage as an unknown, plus the constraint; Method II: KVL around the supermesh, plus the constraint
a dependent source	write its control variable with mesh currents: a constraint equation
node or mesh?	count the equations left; mesh current analysis needs a planar circuit

After the mesh currents are found: $\mathbf{i}=M^T\mathbf{i}_M$, then the branch voltages from the element laws.